

Problem Set 2 – PART 2

ECON 2010 – Macroeconomics

Spring 2019

The entirety of this problem set is **due at the start of class Week 5. YOU HAVE TO DO ALL OF THE PROBLEMS IN THIS ENTIRE PACKET.**

Note that this problem set will work the same as the first problem set and all problem sets. Work on a separate sheet of paper on your own, and we will correct the problem set together in class.

Name:

Problem set instructions.

1. **Write your answers on a separate sheet of paper.** Do not write on this packet until we go over the problem set during class. You will use this problem set sheet to correct the answers that you provide on the separate sheet when we go over the problem set in class the day that it is due.
 2. If possible, please write your final answers in pen.
 3. Please write your name on this packet and on the separate sheet(s) of paper with your answers to the questions.
 4. When appropriate, show your work for each problem. Even if your answer is technically incorrect, you may get many points for the work you did to arrive at that answer. If in doubt, show your work.
 5. Note that not all questions have *one* right answer. Some questions might have many correct answers.
 6. Remember that the purpose of the problems is not to trick you. It is to test your understanding of course material. Don't spend too much time trying to parse little bits of language since more than likely that language is not meaningful.
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Problem 1 (59 points)

This is on part 1 of the problem set handed out last week. Please refer to that handout to complete the problem. Yes, you need to do that problem *in addition to these problems below* by the start of class on week 5.

Problem 2 — The Expenditure Method of Calculating GDP (20 points)

Consider the expenditure method for calculating/accounting Gross Domestic Product (GDP). For each of the statements below, do **both** following.

First, indicate *which* component of US GDP is affected by what happens in the statement. You can answer **one** of the following per statement: none, consumption (C), private investment (I), government purchases (G), or net exports (NX) (you do **not** need to specify imports or exports if you choose net exports) (worth 2 points).

Second, indicate how US GDP is affected by the change described in the statement. You can answer **one** of the following per question: GDP *increases*, GDP *decreases*, GDP *does not change*, or *it is not possible to know how GDP changes given the information provided in the statement* (worth 2 points). Note that this question doesn't ask you specifically how much GDP changes, it is simply asking you about whether it changes, and in what direction — positive, negative, etc. — it changes.

Very important: For each of the statements, use the principle of *all else equal* (a.k.a. ceteris parabis, holding other factors fixed) to conduct your analysis and answer the questions.

- (a) Households in the US become worried about how much water it takes to grow almonds and stop buying and eating US-grown almonds. **(4 points)**

- (b) Claudette, who is a Canadian citizen and who lives in Canada, buys a new car in her home town of Toronto. The car was built by Ford Motors in Michigan. **(4 points)**

- (c) Bridgestone tires purchases a new tire tread stamping machine for its factory outside of Nashville. The steel rolling machine was produced in South Carolina. **(4 points)**

- (d) The city government of Knoxville builds a new 12 mile bike path and park through the city. **(4 points)**

- (e) Kyla quits her job to become a full time caregiver for her sick mother, a job that she would have otherwise paid an in-home nurse to provide. **(4 points)**

Problem 3 — Production in a Magical Economy (13 points)

Consider a magical (and therefore clearly fictional) economy. This magical economy produces two products — lumber, and magical wands. Lumber is used as firewood (a *final good*) to heat homes and cook food because for some reason magic and electricity never go hand in hand. Lumber is also used as an *intermediate good* (the sole/only *input*) in producing magical wands.

Important: It takes 2 pounds of lumber to produce 1 magical wand.

Important: Remember the definition of GDP.

Table 1 below provides the output of lumber, and magic wands, and their prices for two years.

Table 1: Production and prices of lumber and wands

Year	Lumber quantity (lbs)	Lumber price (per lb)	Wand quantity (wands)	Wand price (per wand)
Year 1	80	\$1	15	\$10
Year 2	140	\$2	20	\$15

- (a) Calculate (nominal) GDP for the magical economy in year 1 of Table 1 above. **(4 points)**

(b) Calculate (nominal) GDP for the magical economy in year 2 of Table 1 above. **(4 points)**

(c) Consider the following statement: “In year 2 Dana and Trevor each buy one of the wands that are produced in this magical economy in year 2, but also each own a wand that they bought in year 1 from the wands that were produced in year 1. Dana decides she actually prefers Trevor’s year 1 wand and trades her year 1 wand for his year 1 wand. But after two days, Trevor decides he actually wants both his old year 1 wand and Dana’s year 1 wand and so gives Dana \$8 for his old year 1 wand that he previously traded to her. Now Trevor has his year 2 wand and both his old year 1 wand and Dana’s year 1 wand. After another two days, Dana decides she actually doesn’t want to be a magician anymore and sells her year 2 wand to Trevor for \$14. Now Trevor has all of his wands and Dana’s wands. It cost him \$22 to make the two purchases.”

Answer whether the following statement is **true or false**: “Given the sales and trades described above, GDP in year 2 is the same you found in part (b) above.” In just a few sentences, **explain briefly why the statement is true or false**. Note than an explanation cannot simply be a restatement of the true/false statement. Instead, an explanation draws on things like how GDP is calculated, how trades and sales relate to that definition, etc. **(2 points)**

(d) Answer whether the following statement is **true or false**: “If this magical economy exports **all** of its magic wands to a neighboring magical country in year 2, its GDP in that year is

calculated as zero (\$0).” In just a few sentences, **explain briefly why the statement is true or false**. Note that an explanation cannot simply be a restatement of the true/false statement. Instead, an explanation draws on things like how GDP is calculated, how exports relate to that, etc. **(2 points)**

- (e) “We should use a measure other than GDP to think about human well-being because GDP doesn’t take into account how income/production is distributed, or how healthy people are, or even how happy they are.” Is the above an example of **positive or normative** statement? **(1 point)**

Correcting Your Problem Set (8 points)

You’ll receive 8 points for correcting your problem set in class — that is, marking the answers that you got correct on this sheet, and writing an explanation for why some of your answers were incorrect on others.